

6 (a) State *Faraday's law of electromagnetic induction*.

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 [1]

(b) A long solenoid of length 1.6 m is wound evenly with 400 turns of insulated wire. The solenoid has a circular cross-section of diameter 4.0 cm.

A coil Y having 80 turns of wire is wound tightly around the centre of the solenoid, as illustrated in Fig. 6.1.

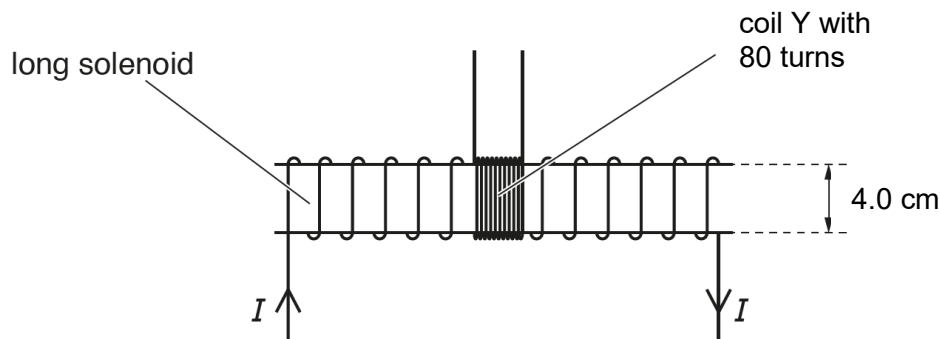


Fig. 6.1

(i) A current I of 3.8 A flows in the solenoid.

Calculate the magnetic flux linkage through coil Y.

magnetic flux linkage = Wb [3]

(ii) The current I in the solenoid in (b)(i) is reversed in 0.30 s.

Calculate the mean e.m.f. induced in coil Y.

mean e.m.f. = V [2]

(iii) The current I in the solenoid in (b)(ii) varies with time t as shown in Fig. 6.2.

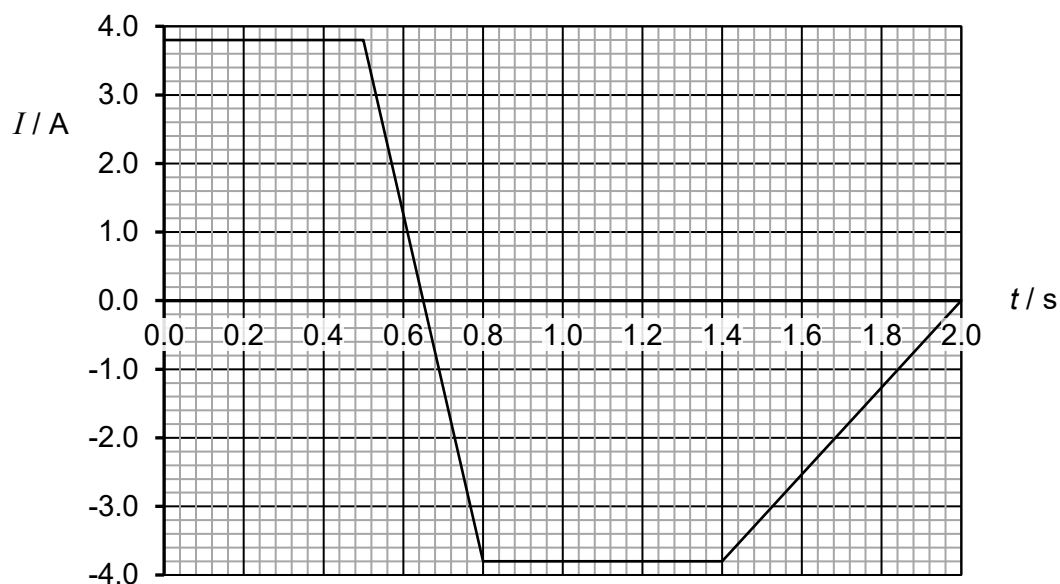


Fig. 6.2

Use your answer to (b)(ii) to sketch, on Fig. 6.3, the variation with time t of the e.m.f. E induced in coil Y.

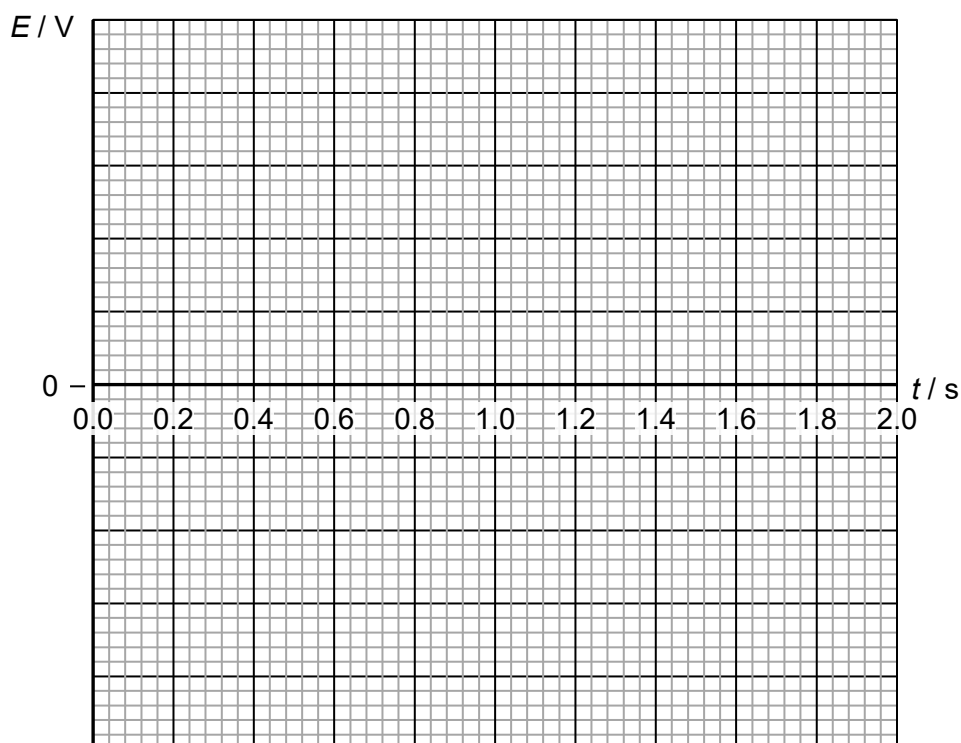


Fig. 6.3

[3]

- (iv) An iron core is inserted into the solenoid and then held stationary within the solenoid.

State and explain the effect, if any, on the change in Fig. 6.3.

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7. (a) The variation with time t of the current I for a sinusoidally alternating voltage supply of